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Comprehensive Review Report on:

Determination of the Cracking Resistance of Hot Mix Asphalt Prepared with Capsules Containing Waste Oil

A. M. Ozdemir, E. Yalcin, M. Yilmaz, B. V. Kok, and A. Garcia Hernandez, Journal of Transportation Engineering, Part B: Pavements, 2024.

1. Introduction and Overview

This manuscript examines the use of calcium alginate capsules filled with waste vegetable oil (WVO) and waste mineral oil (WMO) as a self healing mechanism in hot mix asphalt. Capsule based treatments have gained attention because they allow rejuvenators to be stored inside the mixture and released only when cracking occurs. This reduces the risk of premature softening and creates a targeted healing process that may extend pavement service life. The study also supports sustainability goals by using waste derived oils as rejuvenators.

The authors aim to evaluate how capsule type and dosage influence the mechanical behavior of asphalt mixtures and to determine whether oil released during cracking can restore part of the lost strength. Their work is framed within ongoing efforts to combine durability, healing capability, and environmental benefits in asphalt mixture design. The study provides experimental evidence on how capsules affect stability, fracture resistance, and healing potential, and offers recommended dosage levels that balance healing with acceptable mechanical performance.

2. Methodology

The authors produced calcium alginate capsules through an ionotropic gelation process using sodium alginate solutions and calcium chloride. Capsules were filled with either WVO or WMO, dried, and stored at low temperature to prevent oil loss. Hot mix asphalt specimens were prepared using a standard 50/70

binder and a fixed aggregate gradation. Capsule contents of 0.25, 0.50, 0.75, and 1.00 percent by mixture mass were added without adjusting binder content.

Mechanical performance was evaluated using Marshall stability and flow testing. Fracture resistance was measured through semi circular bending tests. Low temperature tests at 0 C followed linear elastic fracture mechanics principles to calculate fracture toughness, while tests at 25 C followed elastic plastic fracture mechanics to calculate the critical strain energy release rate. Self healing behavior was assessed by fracturing specimens, conditioning them for 120 hours, and retesting to measure strength recovery. FTIR spectroscopy was used to estimate the percentage of capsules that broke during testing by analyzing changes in oil related absorbance peaks.

3. Results

The study found that capsule addition decreased Marshall stability and increased flow, with more severe effects at higher capsule contents. WMO mixtures showed larger reductions in stability than WVO mixtures. Fracture testing showed that low capsule contents, particularly around 0.25 percent, produced slight increases in fracture toughness at low temperature, but higher contents reduced both toughness and strain energy release rate.

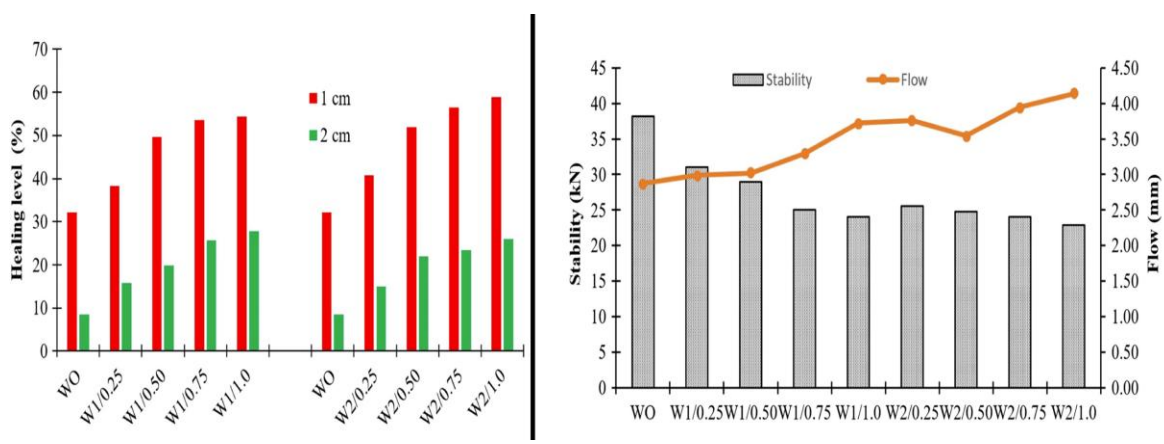

Figure 1: Mechanical strength decreases with capsule content while healing ability increases, demonstrating the fundamental tradeoff in capsule-based self healing asphalt mixtures.

Healing tests showed clear improvements in strength recovery when capsules were included. Healing ratios increased with capsule content, and WMO mixtures generally healed more than WVO mixtures because they released more oil.

Specimens fractured at 0 C also healed more effectively than those fractured at 25 C. FTIR analysis confirmed that higher capsule contents and higher temperatures led to greater oil release, which aligned with the observed healing behavior.

The authors conclude that low capsule dosages can enhance healing while keeping mechanical performance within acceptable limits. They identify 0.25 percent as the most effective balance between healing capability and structural performance.

4. Detailed Evaluation of the Study

The manuscript is clearly structured and follows a logical sequence from materials through testing to interpretation. The capsule production process is explained in sufficient detail to be repeatable, and the use of identical binder contents across mixtures isolates capsule effects. The authors also distinguish between mineral and vegetable oil capsules, enabling a more nuanced comparison of how oil chemistry influences both mechanical response and healing.

However, several fundamental issues limit the strength of the conclusions. The most significant is the absence of statistical indicators such as standard deviation or confidence intervals. Asphalt mixtures are inherently variable, and capsule addition introduces additional heterogeneity. Presenting only mean values for stability, flow, fracture toughness, or healing obscures the true reliability of the reported trends. Without statistical treatment, the differences between mixtures cannot be confidently attributed to capsule presence rather than normal scatter.

Another major limitation is the incomplete justification of the capsule dosage range. The study uses four levels from 0.25 to 1.00 percent by total mixture mass, yet provides no discussion regarding the practical or theoretical basis for choosing this spectrum. The upper levels result in severe declines in stability and Marshall quotient, raising questions about whether these dosages would ever be considered realistic. A stronger conceptual framework for dosage selection would have improved the study's engineering relevance. The interpretation of Marshall results would also benefit from deeper discussion. Stability reductions of over 30 percent in some mixtures are significant and would likely impact rutting potential and early life performance. Similarly, flow values increased to levels beyond specification limits in the highest WMO content mixtures. The study describes these results but

does not critically assess their implications for pavement design or possible mix adjustments to mitigate the negative impacts of capsule inclusion.

The fracture mechanics section demonstrates thoughtful selection of test temperatures and loading regimes, but the assumptions of LEFM and EPFM are not justified explicitly. More information regarding the linearity of the load deflection response before failure and the validity of KIC calculations for asphalt at low temperatures would strengthen confidence in the results. The procedure for estimating the derivative of strain energy with respect to notch depth in the EPFM portion is mentioned only briefly and needs more elaboration to ensure reproducibility.

5. Assessment of the Self Healing Component

The investigation of healing behavior is one of the strongest aspects of the paper. The authors use a consistent protocol involving complete fracture, conditioning at controlled temperature for 120 hours, and retesting. Although complete fracture does not fully represent field damage, the method provides a repeatable basis for comparing healing efficiency. The finding that healing is more pronounced in mixtures containing WMO capsules aligns with the FTIR results showing greater oil release from WMO based capsules.

Nevertheless, the authors do not discuss whether the healing levels observed under controlled laboratory conditions would translate meaningfully into real pavements. The 120 hour rest period at 25 C bears little resemblance to realistic thermal cycles, traffic loads, or crack widths. A more robust analysis would compare the healing ratios obtained here with those from partial damage tests, repeated healing cycles, or environmental conditioning to evaluate long term performance.

The FTIR methodology is creative and adds scientific depth to the paper, but its description is too brief. The relationship between absorbance at 1745 cm minus one and oil concentration is treated as linear without supporting calibration data. The reference spectrum representing one hundred percent broken capsules is not described in detail, leaving ambiguity in how the percentage of oil release was quantified. Despite these limitations, the FTIR data support the authors' interpretation that higher capsule content results in more breakage and more released oil.

6. Strengths of the Research

The study addresses a timely and meaningful topic at the intersection of pavement materials engineering and sustainability. Its investigation of waste oils within a controlled encapsulation system distinguishes it from more conventional rejuvenator studies. The combination of mechanical, fracture, and spectroscopic analysis provides a multi dimensional understanding of capsule behavior. The identification of an optimal low dosage level for balancing healing capability and mechanical performance is a valuable insight for future self healing mixture design.

7. Weaknesses and Areas Needing Improvement

The most significant weaknesses lie in the areas of experimental justification, statistical rigor, and engineering interpretation. The study would benefit from:

- More robust justification for capsule dosages based on expected field performance or prior mechanistic modeling.
- Inclusion of statistical measures to quantify variability and support conclusions.
- Deeper discussion on how the observed reductions in Marshall properties influence expected pavement performance.
- Clearer explanation of fracture mechanics assumptions and calculations.
- More explicit comparison between laboratory healing and realistic field healing scenarios.

These revisions would greatly enhance the credibility and engineering utility of the findings.

8. Contribution to Pavement Engineering Practice

Despite its limitations, the paper contributes meaningfully by demonstrating that encapsulated waste oils can provide measurable healing benefits while influencing mechanical behavior. The distinction between WVO and WMO mixtures highlights the importance of rejuvenator chemistry. This work also lays

groundwork for future optimization efforts in mixture design involving capsules. If refined, such systems could support longer lasting pavements, reduced maintenance interventions, and better resource utilization.

9. Final Recommendation

The manuscript has merit and presents novel results, but substantial clarification and additional analysis are required before it can be fully relied upon for engineering decisions. I would classify my recommendation as Major Revision. The authors should strengthen the justification of methods, incorporate statistical analysis, and discuss practical implications more clearly. With these improvements, the paper could become a strong contribution to the field of self healing asphalt materials.

References:

Ozdemir, A. M., Yalcin, E., Yılmaz, M., Kok, B. V., & Hernandez, A. G. (2024). Determination of the cracking resistance of hot-mix asphalt prepared with capsules containing waste oil. *Journal of Transportation Engineering, Part B: Pavements, 150*(4), 04024042. https://doi.org/10.1061/JPEODX.PVENG-1315